



毕 业 论 文

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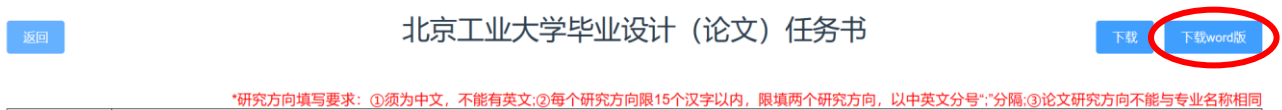
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摘 要

摘要是论文内容的简要陈述，应尽量反映论文的主要信息，内容包括研究目的、方法、成果和结论，不含图表，不加注释，具有独立性和完整性。中文摘要一般为 400 字左右，英文摘要应与中文摘要内容完全相同。“摘要”字样位置居中。关键词是反映毕业设计（论文）主题内容的名词，是供检索使用的。主题词条应为通用技术词汇，不得自造关键词，尽量从《汉语主题词表》中选用。关键词一般为 3-5 个，按词条外延层次（学科目录分类），由高至低顺序排列。关键词排在摘要正文部分下方。

关键词：关键词 1；关键词 2；xxx

Abstract

The abstract is a succinct statement of the content of the thesis, designed to reflect the main information of the paper as fully as possible. It includes the purpose of the research, methods, results, and conclusions, without containing any figures, tables, or annotations. The abstract should be independent and complete. The length of the Chinese abstract is generally around 400 characters, and the English abstract should contain exactly the same content as the Chinese one. The word "Abstract" should be centered.

Keywords are nouns that reflect the theme of the graduation project (thesis) and are used for indexing purposes. Keywords should be general technical terms and should not be coined by the author, preferably selected from the "Chinese Thesaurus". Typically, there are 3-5 keywords, arranged in order of their scope (according to the classification of the subject directory) from broad to specific. The keywords are placed below the main text of the abstract.

Keywords: keyword1, keyword2 , xxx

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1. 介绍和使用

欢迎使用北京工业大学本科生毕业设计（论文） $\text{\LaTeX}$ 模版。本模版是基于往届学长开源的北京工业大学研究生学位论文模板制作完成的，主体架构照搬了研究生论文的模板，作为乘凉的后人，我在此满怀感激地向前人的工作而努力致以诚挚的敬意。

基于“北京工业大学本科生毕业设计（论文）撰写规范”，本模版对研究生模板的主要修改在于封面、行距、页边距、字体、“毕业设计（论文）诚信声明书”以及“关于论文使用授权的说明”的内容。

请使用**XeLaTeX**编译器编译**main.tex**文件。请务必按照自己所在年份的具体要求撰写论文，如需微调模板具体设置，请修改**bjutthesis.cls**的相关内容。

2024 年 5 月 9 日更新：document 文件夹提供了 2024 年信息学部毕业设计管理系统发布的若干文件，在这里提醒所有使用该模板的同学注意查看，**比对并确认自己所在年份是否有特殊要求，请务必按要求进行论文撰写**，以免造成不必要的麻烦。

1.1 如何使用

1.1.1 模版文件总览

首先要明确的是， $\text{\LaTeX}$ 出于安全的考虑，在 unix 环境下，往上一级目录写入编译时文件（如 aux 文件）时会被禁止。为了最大兼容性，当前版本的模版将所有的文件都放置在同一级目录下。（不排除某些 tex 的特殊编译设置或单独设置的环境变量）。**图片可以放在二级目录，例如新建 pics 文件夹，使用“bjutthesis/pics/XXX.pdf”添加图片。**

大多数用户使用时，都不必关注所有文件，只需了解如下几个文件：

主文档 : main.tex	中文摘要 : cabstract.tex
英文摘要 : eabstract.tex	章节文档 : chapt*.tex
引用文档 : reference.bib	用户出版物文档 : publication.tex
致谢文档 : acknowledge.tex	

以上文档使用时，基本上只需填入文字内容即可，不需要考虑格式。

1.1.1.1 主文档主文档正文部分组织了论文结构，论文正文内容由chapt1.tex, chapt2.tex等组织。用户自行添加，参考章节文档一节。

1.1.1.2 中文摘要中文摘要内容由cabstract环境包裹，两个段落之间以大于 1 行的空行作为分割。

在中文文档中，有ckeywords环境包裹的关键词，以”；”分割。

1.1.1.3 英文摘要英文摘要与中文摘要一致，由eabstract环境包裹。

类似的，也有ekeywords环境包裹的关键词，但以”，”分割。

1.1.1.4 章节文档章节文档是论文的主要内容，在章节文档中，包含不同的小结，本模版共4级小结，分别由bisection, subsection, subsubsection, subsubsubsection标记。其中，目录会记录至第二级小结，即subsection。具体书写格式参考本文档源码(chapt1.tex)。

1.1.1.5 引用文档引用文档格式为正常bib file。你可以使用cite{}进行上标引用，如：

RSA\cite{rsa} 算法

也可以使用citen{}进行正文引用，如：

文献\citen{rsa} 提出了...

1.1.1.6 出版物文档直接在publication环境中添加即可。

1.1.1.7 致谢文档同引用文档。

1.2 基本语法

面介绍一些基本语法。可自行修改。

1.2.1 如何插入图片并引用

```
\begin{figure}[htpb]
  \centering
  \includegraphics[width=0.8\textwidth]{bjut_logo_color.pdf}
  \caption{中文图题}
  \label{fig:logo}
\end{figure}
```

% 使用\cref{<label>} 引用：如\cref{fig:logo} 所示



图 1.1 中文图题

“如图 1.1 所示”

1.2.2 如何插入表格并引用

```

\begin{table}[htpb]
\centering
\caption[] {三线表} {A Table Example}
\label{tab:1st}
\begin{tabular}{@{}llr@{}}
\toprule
\multicolumn{2}{c}{Item} & \\
\cmidrule(r){1-2}
Animal & Description & Price(\$) \\
\midrule
Gnat & pergram & 13.65 \\
& each & 0.01 \\
Gnu & stuffed & 92.50 \\
Emu & stuffed & 33.33 \\
Armadillo & frozen & 8.99 \\
\bottomrule
\end{tabular}
\end{table}
% 引用方法与图片一致 如\cref{tab:1st} 所示

```

表 1.1 三线表

Item		
Animal	Description	Price(\$)
Gnat	pergram	13.65
	each	0.01
Gnu	stuffed	92.50
Emu	stuffed	33.33
Armadillo	frozen	8.99

“如表 1.1 所示”

1.2.3 如何插入公式

```

\begin{equation}\label{eq:eq1}
e^{\pi i} + 1 = 0
\end{equation}

```

$$e^{\pi i} + 1 = 0 \quad (1.1)$$

1.2.4 如何插入算法

```
\begin{algorithm}[htbp]
  \caption{1+1 等于几}
  \label{alg:algorithm}
  \begin{algorithmic}[1]
    \REQUIRE $1$.
    \ENSURE $1$.
    \STATE $i \leftarrow 1$
    \FOR {$i \leq 1$}
      \STATE $i \leftarrow i + 1$;
    \ENDFOR
    \STATE Return $i$.
  \end{algorithmic}
\end{algorithm}
```

算法 1-1: 1+1 等于几

输入： 1.

输出： 1.

- 1: $i \leftarrow 1$
 - 2: **for** $i \leq 1$ **do**
 - 3: $i \leftarrow i + 1$;
 - 4: **end for**
 - 5: Return i .
-

结论

结论是毕业设计（论文）的总结，是整篇论文的归宿，要求精炼、明确、完整，能准确表达毕业设计（论文）的成果。

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